



VIT-AP
UNIVERSITY

Module 1: Understanding Cloud Computing

Cloud origins and influences

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OUTLINE

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 - **Development of the Internet**
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- Cloud computing:
 - Delivery of various services over the internet.
 - Services include:
 - Storage
 - Databases Servers
 - Networking
 - Software
 - Analytics Intelligence
 - Key benefits:
 - Faster innovation
 - Flexible resources
 - Economies of scale

Early Concepts and Precursors

- Early Concepts and Precursors
 - Mainframe and Time-sharing (1950s-1960s).
 - Large, powerful computers primarily used by businesses and government organizations.
 - Capabilities: Capable of handling and processing vast amounts of data.
 - Usage: Centralized computing; multiple users connected via terminals.
 - Cost: Expensive and required significant investment in infrastructure.
 - Time-Sharing:
 - Concept: Multiple users accessing a single mainframe computer simultaneously.
 - Technology: Use of terminals connected to the mainframe.
 - Efficiency: Improved resource utilization by sharing computing power among many users.
 - Innovation: Allowed multiple users to interact with a computer at the same time, paving the way for the development of modern multi-user systems.
 - Impact: Laid the groundwork for later developments in distributed computing and cloud computing by demonstrating the feasibility of resource sharing.
 - Significance:
 - Resource Sharing: Demonstrated the benefits and efficiency of sharing computing resources.
 - User Experience: Enhanced user experience by allowing multiple users to run applications concurrently.

Early Concepts and Precursors

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 - ARPANET (1960s-1970s).
 - ARPANET: Advanced Research Projects Agency Network. Initiated by: U.S. Department of Defense's ARPA (Advanced Research Projects Agency).
 - UCLA (University of California, Los Angeles) and ARPANET
 - Impact on Cloud Computing:
 - Networking Foundation: ARPANET's success showcased the viability of large-scale, distributed networks, fundamental to cloud computing.
 - Ongoing Influence: Packet switching and network protocols developed during ARPANET's early days continue to underpin cloud infrastructure today.

Development of the Internet

- Development of the Internet
 - Evolution of the Internet (1980s-1990s) & Growth of networking and TCP/IP protocol.
 - 1980's TCP/IP Protocol
 - 1983, foundational protocol for the internet
 - 1984, DNS (like www.example.com) into IP addresses,
 - Email and Newsgroups :
 - Communication, forum for discussions on various topics
 - 1990s: World Wide Web Commercialization
 - HTTP (Hypertext Transfer Protocol) and HTML (Hypertext Markup Language) enabled the creation and linking of web pages

- ❑ Introduction
- ❑ Early Concepts and Precursors
- ❑ Development of the Internet

Development of the Internet

- Development of the Internet

- Web browsers:

- Mosaic graphical web browser for non technical users), released in 1993 Netscape Navigator (popularized web browsing), launched in 1994

- Commercialization:

- Search Engines:

- Early search engines like Archie, Gopher, and later, Yahoo! (1994) and Google (1998),

- helped users find information on the growing web.

- E-commerce:

- Introduction of online shopping

Virtualization Technology

- Virtualization Technology

- Introduction of Virtual Machines (VMs) & Impact of Virtualization.

- VMs are software-based emulations of physical computers, allowing multiple VMs to run on a single physical machine
 - VMs replicate the architecture of a real computer, including CPU, memory, storage, and networking. Allows multiple operating systems to run on a single hardware platform, isolated from each other.
 - Virtualization is a cornerstone of cloud computing, enabling multi-tenancy and resource pooling.
 - Increased Efficiency, Cost Savings, Dynamic Allocation, Scalability

- ❑ Introduction
- ❑ Early Concepts and Precursors
- ❑ Development of the Internet
- ❑ Virtualization Technology

- Grid Computing
 - Concept and Development, Projects and Examples.
 - Grid computing is a distributed computing model that involves a network of **loosely coupled computers** working together to perform tasks, often involving complex computations or large datasets.
- Characteristics:
 - Distributed Resources, Parallel Processing, Resource Sharing, Scalability, Heterogeneity, Virtual Organizations
 - via grid technologies.
 - Scientific Research Projects:
 - » SETI@home: in 1999. Volunteers use spare processing power on their personal computers to analyze radio signals
 - Environmental and Climate Research:
 - » Earth System Grid: studying global climate change.

Birth of Cloud Computing

- Birth of Cloud Computing

- Early Cloud Services, Key Milestones.

- Cloud computing evolved from the development and convergence of internet services and distributed computing technologies.
 - 1960s: IBM's mainframes offering time-sharing services.
 - 1990s:Salesforce.com (1999): delivering applications over the internet.
 - 2002:Amazon Web Services -Public.
 - 2006: Google Cloud Platform (GCP) launched, including Google App Engine for web applications.
 - 2008:Microsoft Azure , 2009: VMware , 2010:OpenStack: An open-source cloud computing platform, 2014:Docker popularized containerization, improving cloud computing efficiency. 2019:Kubernetes: Open-sourced by Google, became the standard for container orchestration in the cloud.

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Influences on Cloud Computing

- Influences on Cloud Computing

- Technological Advances.

- Virtualization, Broadband Internet, Distributed Computing, Containerization

- Economic Factors.

- Cost Reduction, Pay-as-You-Go Pricing, Resource Optimization

- Business Needs.

- Scalability,
 - Disaster Recovery and Business Continuity,
 - Remote Work Enablement,
 - Collaboration

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- ❑ Impact, Challenges & Future

Impact, Challenges & Future

- Impact of Cloud Computing
 - On Businesses:
 - Innovation, scalability, reduced IT overhead.
 - On Individuals: Accessibility, collaboration, remote work capabilities.
- Challenges and Considerations
 - Security and Privacy:
 - Data breaches, compliance.
 - Reliability and Downtime:
 - Service outages, dependency on providers.
- Future of Cloud Computing
 - Emerging Trends:
 - Edge computing, AI integration, multi-cloud strategies.
 - Potential Developments:
 - Increased automation, advanced cybersecurity measures.

Summary on Cloud origins and influences

❑ Introduction

- ❑ Cloud computing
 - ❑ services
 - ❑ Key benefits

❑ Early Concepts and Precursors

- ❑ Mainframe and Time-sharing (1950s-1960s).
- ❑ ARPANET (1960s-1970s).

❑ Development of the Internet

- ❑ Evolution of the Internet (1980s-1990s).
- ❑ Growth of networking and TCP/IP protocol

❑ Virtualization Technology

- ❑ Introduction of Virtual Machines (VMs).
- ❑ Impact of Virtualization.

❑ Grid Computing

- ❑ Concept and Development.
- ❑ Projects and Examples.

❑ Birth of Cloud Computing

- ❑ Early Cloud Services.
- ❑ Key Milestones.

❑ Influences on Cloud Computing

- ❑ Technological Advances.
- ❑ Economic Factors.
- ❑ Business Needs.

Quiz

- **Module 1: Cloud Origins and Influences**
- **Q1: What was one of the earliest concepts that contributed to the development of cloud computing?**
 - A) Mainframe and Time-sharing
 - B) Personal Computers
 - C) Smartphones
 - D) Blockchain
 - **Answer:** A) Mainframe and Time-sharing
- **Q2: Which project is considered a precursor to the internet and influenced the development of cloud computing?**
 - A) Windows NT
 - B) ARPANET
 - C) Linux
 - D) SETI@home
 - **Answer:** B) ARPANET
- **Q3: Which company launched the first significant cloud service, Amazon Web Services (AWS)?**
 - A) Microsoft
 - B) Google
 - C) Amazon
 - D) IBM
- **Discuss the historical evolution of cloud computing, starting from the early concepts of mainframe and time-sharing, through the development of ARPANET, to the introduction of virtualization and grid computing. How did these advancements contribute to the creation of modern cloud computing?**

References

1. Thomas Erl, Ricardo Puttini, Zaigham Mahmood, “Cloud Computing: Concepts, Technology & Architecture”, PHI Publications, 2013.

Questions



**Thank
You**